

Ananya Joseph

Machine Learning Engineer

Dallas, Tx

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Professional Summary

Master's graduate in Applied Statistics and Data Science with 3+ years of industry experience and proficiency in PyTorch, TensorFlow, Keras, and deep learning frameworks. Developed practical ML expertise through diverse roles including internship, graduate research assistant, and student technical assistant specializing in data science and visualization. Ready to drive impactful data-driven solutions.

Experience

Dallas College

Dallas, Texas

Data Analyst

12/2025 - Present

- **Data Analytics & Institutional Impact:** Achieved 15%+ efficiency gains in course scheduling, resource allocation and enhanced student retention strategies, measured through interactive Power BI dashboards and evidence-based policy adoption, by conducting multivariate statistical analysis using SAS, SQL, and Python.
- **Process Automation:** Achieved 30% reduction in manual reporting time and ensured real-time data accuracy, measured by streamlined institutional decision-making, through automated Python and SQL pipeline engineering.

University of Texas at Arlington

Arlington, Texas

Graduate Research Assistant

08/2025 - 12/2025

- **Automated Data Architecture & Insights:** Engineered **R and SQL** scripts to automate student performance data storage, transitioning raw data into a structured database; developed a custom classification pipeline for qualitative student survey data, transforming unstructured feedback into actionable tabular insights and **Power BI** dashboards.
- **Advanced Predictive Modeling & Policy Impact:** Applied **statistical modeling and machine learning** (Python/SAS) to identify success drivers, improving research precision by 15%; these insights enabled data-driven policy changes and targeted support programs that resulted in a **10% increase in student satisfaction scores**.
- **Engineered Secure AI Architectures:** Spearheaded research into **Graph Neural Networks** to identify link-stealing vulnerabilities within the automotive domain; utilized **PyTorch and TensorFlow** to develop deep learning-based threat models and defensive countermeasures, significantly enhancing model resilience against adversarial attacks.

Office of Information Technology, University of Texas at Arlington

Arlington, Texas

Student Technical Assistant

11/2024 - 08/2025

- **Engineered multi-source data pipelines** by integrating **ServiceNow, SiteImprove, and Excel** into centralized **Power BI** dashboards; automated the tracking of Key Performance Indicators that improved project health metrics by **15%** over a 12-month period through enhanced visibility and resource allocation.

Glitz IT Solutions

Kerala, India

Software Engineer

11/2021 - 05/2024

- **API development** and Collaborated with senior engineers to diagnose and rectify bottlenecks in **database architecture**, leading to a demonstrable **10%** enhancement in application responsiveness and processing efficiency

Education

University of Texas at Arlington

Arlington, Texas

Master of Science in Applied Statistics and Data Science

08/2024 - Present

Cochin University of Science and Technology

Cochin, India

Bachelor of Technology in Information Technology

08/2016 - 10/2020

Skills

- Python, R, SAS code, C##, C++
- Apache Spark, Apache Flink, Apache Kafka
- Polars, Arrow, Pandas, NumPy SAS
- Excel, Power BI, Tableau, Databricks, Snowflake
- Visual Studio Code, Jupyter Notebook, Git, Postman, SAS, R Studio
- Docker, Putty, Kubernetes
- PyTorch, TensorFlow, Keras, Scikit-learn
- Crew AI, LangChain, LangGraph
- SQL, ggplot2, Matplotlib, Seaborn
- Duckdb, PostgreSQL, MySQL
- Cursor, Antigravity, NotebookLM
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Certifications

Databricks: Certified Data Analyst Associate

AWS: Generative AI, Machine Learning

Snowflake: Ecosystem and Architecture

LangChain: LLM Application Development, Langgraph: AI Agents